

Proof Workout Mastery Workbook

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I have neither given nor received unauthorized assistance.

Warm Up THM 1 (Given Example)

Prove, for all positive integers, A, B , if A and B are even, then $C = A + B$ is even:

There is d such that $A = 2d$ definition of even

There is h such that $B = 2h$ definition of even (*notice d/h cannot be the same, why?*)

$C = A + B$ premise

$= 2d + 2h$ substitution

$= 2(d + h)$ factoring

Define k integer as $k = d + h$ closure of the integers, renaming $d + h$

$= 2(k)$ substitution

- Thus we see that $C = 2(k)$, k is an integer, so by definition of even, C is even.

□

Use Warm Up THMs 2–4 for ungraded practice.

Warm Up THM 2

For all positive integers, A , B , if A is even and B is odd, $A + B = C$ is odd.

There is d such that $A = 2d$ definition of even

There is h such that $B = 2h + 1$ definition of odd

$C = A + B$ premise

$= 2d + (2h + 1)$ substitution

$= 2d + 2h + 1$ algebra

$= 2(d + h) + 1$ factoring

Define k integer as $k = d + h$ closure of the integers, renaming $d + h$

$= 2(k) + 1$ substitution

- Thus we see that $C = 2(k) + 1$, k is an integer, so by definition of odd, C is odd.

□

Warm Up THM 3

For all positive integers, A , B , if A is odd and B is odd, $A + B = C$ is even.

Let $\mathbb{Z}$ be the set of all integers, then:

Let $A = 2x + 1$ such that $x \in \mathbb{Z}_{\geq 0}$ definition of odd

Let $B = 2y + 1$ such that $y \in \mathbb{Z}_{\geq 0}$ definition of odd

$C = A + B$ premise

$= (2x + 1) + (2y + 1)$ substitution

$= 2x + 2y + 2$ algebra

$= 2(x + y + 1)$ factoring

Let z be an integer $z = x + y + 1$ closure of integers

$= 2(z)$ substitution

- Thus we have shown that $C = 2(z)$ such that $z \in \mathbb{Z}_{\geq 0}$, so by definition C is even.

□

Warm Up THM 4

Prove for all positive integers, A , B , if A and B are even, $A \cdot B = C$ is even.

Warm Up THM 5

Prove: For all positive integers, A , B , if A is even and B is odd, $A \cdot B = C$ is even.

#1 (Graded) – Warm Up THM 6

Prove: For all positive integers, A , B , if A and B are odd, $A \cdot B = C$ is **odd**.

We will now prove that $A \cdot B = C$ is odd.

There exists an integer j such that $A = 2j + 1$ definition of odd

There exists an integer k such that $B = 2k + 1$ definition of odd

$C = A \cdot B$ premise

$= (2j + 1) \cdot (2k + 1)$ substitution

$= 4jk + 2j + 2k + 1$ FOIL

$= 2(2jk + j + k) + 1$ factoring

Let h be an integer such that $h = 2jk + j + k$ closure of the integers
(under multiplication and addition)

$= 2(h) + 1$ substitution

- Thus we have shown that $C = 2h + 1$, therefore by the definition of an odd integer, C is an odd integer.

□

#2 Prove that if x is odd, then $5x + 3$ is “...” two different ways.

- A direct proof by **applying the Warm-Up Thms** – this means just USE the RESULTS of what you have proved above.

We will now prove that if x is an odd integer, then $5x + 3$ is **even**.

Let $5x = A$, then we know A is an odd integer. We deduce this from Theorem 6 which states if A and B are odd integers, then $A \cdot B = C$ is also an odd integer.

Let B be an odd integer such that $B = 3$. By definition, B is odd because 3 is an odd number.

$5x + 3 = A + B$ by substitution

- By Theorem 3, the sum of two odd integers is even. Thus, we have shown that if x is odd, then $5x + 3$ is even. $\square$

- Using the *techniques* of the warm up Thm. and the formal definitions of even and odd. (This will look a lot like your WUT proofs.)

We will now prove that if x is an odd integer, then $5x + 3$ is **even**.

Let $x = 2m + 1$, where $m \in \mathbb{Z}_{\geq 0}$ definition of an odd number

$5x + 3 = 5(2m + 1) + 3$ by substitution

$= 10m + 5 + 3$ distribution/multiplication

$= 10m + 8$ addition

$= 2(5m + 4)$ factoring

Let n be an integer greater than zero such that $n = 5m + 4$ closure of the integers (under multiplication and addition)

$= 2(n)$ by substitution

- Thus, $5x + 3 = 2n$ when n is an integer greater than zero. Therefore by definition, $5x + 3$ is even. $\square$

Ungraded reflection: Explain the difference between using the RESULTS of a Theorem and using the TECHNIQUES of a theorem's proof.

You may NOT use Warm-Up THM results again unless specified – this includes exams.

Ungraded: Attempt to use a direct proof (similar to techniques of WUT THMs):

If $3n + 7$ is odd, then n is even (n is an integer).

Why is this not working? (hint: it should not work)

This is a good one to discuss on Piazza.

#3 Prove using a proof by contrapositive. (Be sure to use the right format.)

If $3n + 7$ is odd, then n is even (n is an integer).

We will prove this by contrapositive. Lets assume that n is odd, then we will prove $3n + 7$ is even.

Let $n = 2a + 1$ such that a is a positive integer. By the definition of an odd number

$$\begin{aligned} 3n + 7 &= 3(2a + 1) + 7 && \text{by substitution} \\ &= 6a + 3 + 7 && \text{by distribution and multiplication} \\ &= 6a + 10 && \text{by addition} \\ &= 2(3a + 5) && \text{factoring} \end{aligned}$$

$$\begin{aligned} \text{Let } b &= 3a + 5, \text{ then } b \text{ is an integer} && \text{closure of the integers} \\ &&& \text{(under multiplication and addition)} \\ &= 2(b) && \text{by substitution} \end{aligned}$$

Thus, we have shown that if n is odd, then $3n + 7$ is even. Then by contrapositive, If $3n + 7$ is odd, then n is even.

□

Why does this work, when a direct proof did not?

In this case, a direct proof would not allow us to begin with the clean definition of an odd or even integer. This leaves us with a poor setup to proving n is even, and we get stuck in the middle of the proof. Starting with $3n + 7 = 2k + 1$ gives you $n = \frac{2k-6}{3}$, and dividing by 3 makes it impossible to cleanly factor out a 2 to show n is even. Since the definition of a direct proof is $p \rightarrow q$ where p is the expression " $3n + 7$ is odd" and q is " n is even", a proof by contrapositive ($\neg q \rightarrow \neg p$) allows us to flip the expression, which includes the starting point of our proof to " n is odd", allowing us to start with a definition making the proof much easier.

#4 Prove for positive integers, if n^2 is even, then n is even. ⟨careful, careful, careful⟩

We will again prove this by contrapositive. Assume n is odd.

Let $n = 2a + 1$ such that a is a positive integer. By the definition of an odd number

$$n^2 = (2a + 1)^2 \quad \text{using substitution}$$

$$= 4a^2 + 4a + 1 \quad \text{FOIL}$$

$$= 2(2a^2 + 2a) + 1 \quad \text{factoring 2 out}$$

$$= 2(2a(a + 1)) + 1 \quad \text{factoring } a \text{ out}$$

Let $b = a + 1$ then b is an integer closure of the integers
(under addition)

Let $c = 2a(a + 1) = 2ab$, then c is an integer closure of the integers under multiplication
since we have shown b is an integer

$$= 2(c) + 1 \quad \text{by substitution}$$

Thus we have shown that $n^2 = 2(c) + 1$ and therefore by definition n^2 is odd. By contrapositive, since we have shown that if n is odd, then n^2 is odd, then if n^2 is even, n is also even.

□

#5 Prove that $13n + 3$ is even if and only if n is odd. n is an integer. (Must prove both directions, why?)

The above statement follows the same logic pattern as a proof of equivalence. Proofs of equivalence follow the tautology $(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$. So, in order to prove the statement above, we must prove both direction $p \rightarrow q$ and $q \rightarrow p$ where p is " $13n + 3$ is even" and q is " n is odd".

Direction 1: If n is odd, then $13n + 3$ is even.

We will prove this statement directly.

Let $n = 2q + 1$ such that q is a positive integer. By the definition of an odd number

$$\begin{aligned} 13n + 3 &= 13(2q + 1) + 3 && \text{by substitution} \\ &= 26q + 13 + 3 && \text{by distributive multiplication} \\ &= 26q + 16 && \text{by addition} \\ &= 2(13q + 8) && \text{factoring 2} \end{aligned}$$

Let $p = 13q + 8$ then p is an integer closure of the integers
(under addition)

$$= 2(p) \quad \text{by substitution}$$

Thus, we have shown that $13n + 3 = 2p$ and therefore by definition, $13n + 3$ is even.

Direction 2: If $13n + 3$ is even, then n is odd.

We will prove this by contraposition. Assuming n is even, we have

Let $n = 2r$ such that r is a positive integer. By the definition of an even number

$$\begin{aligned} 13n + 3 &= 13(2r) + 3 && \text{By substitution} \\ &= 26r + 3 && \text{By multiplication} \\ &= 26r + (1 + 2) && \text{By substitution } (1 + 2 = 3) \\ &= (26r + 2) + 1 && \text{By associativity} \\ &= 2(13r + 1) + 1 && \text{factoring} \end{aligned}$$

Let $s = 13r + 1$ then s is an integer closure of the integers
(under addition)

$$= 2(s) + 1 \quad \text{by substitution}$$

Thus, we have shown that $13n + 3 = 2s + 1$ and is therefore by definition odd. So, since we have shown that if n is even then $13n + 3$ is odd, then by contraposition if $13n + 3$ is even, then n is odd.

Since we have shown both **Direction 1** and **Direction 2** are true, then we have proven the equivalence " $13n + 3$ is even if and only if n is odd".

□

Ungraded – try this in your own words.

Prove there is no largest integer z . Assume the domain of integers.

There is no integer z , for all n , $z > n$.

Use a proof by contradiction. Review the directions for a proof by contradiction. After your proof briefly discuss – What does this imply about the nature of infinity? Can infinity be an integer?

Say “We prove by...”

For this, we will prove by contradiction.

State the negation –

Assume that there exists an integer z , such that for all n , $z > n$ where z and n are both integers.

Consider the negation.

Assuming the above statement to be true, we expect when proving that a step will be logically contradictory.

Do something valid with the negation that shows something goes wrong:

If this is true, then:

Let $n = z + 1$ if z is an integer then so is $z + 1$

$z \not> (z + 1)$ substitution

State “Contradiction!!!”

But we know that by the definition of integers and addition, $z < z + 1$!

State “by contradiction we have...”

Thus, we have a contradiction because by the laws of addition and integers $z > (z + 1)$ must be false.

What does this imply about the nature of infinity? Can infinity be an integer?

This implies that infinity is not an integer! Which is pretty strange to think about but it makes sense. While infinity does not exist this does make sense in terms of the laws of the set of integers. Essentially if you try to pick a z that is greater than all n , you will never find z because there is always an n that's greater, meaning the set is unbounded and infinity does not exist.

#6 Assume the domain of positive integers. Prove if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Prove by contradiction (using required steps).

We will prove this by contradiction. Assume for contradiction that there exists an $n = ab$, $\neg(a \leq \sqrt{n}) \equiv a > \sqrt{n}$ and $\neg(b \leq \sqrt{n}) \equiv b > \sqrt{n}$ (by DeMorgan's law). Then if $a > \sqrt{n}$ and $b > \sqrt{n}$:

Let $a > \sqrt{n}$	As assumed in our initial statement
Let $b > \sqrt{n}$	As assumed in our initial statement
$ab > \sqrt{n} \cdot \sqrt{n}$	In this case because $a, b, n > 0$, by laws of multiplication this statement is true
$> n$	by multiplication laws

But this leads to a contradiction, because we assumed $n = ab$ but here, $ab > n$! Thus, we have shown there does not exist an $n = ab$ such that $a > \sqrt{n}$ and $b > \sqrt{n}$, and therefore if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ is true by contradiction.

□

#7 We define an exponent b of a , (a is an element of the positive real numbers, and b is an element of the natural numbers) as:

a^b is a multiplied by itself b times. Example $a^4 = a \cdot a \cdot a \cdot a$
 $a^0 = 1$

We will use only the basic principles of algebra (associativity, commutativity, etc.), and the definition of exponents given above to prove the rules of exponents.

Assume a and b are elements of the positive real numbers. You must explain each step. Notice the format of this example proof and follow this structure for #8 and #9. Do not “work both sides of the equation.”

Notice how this proof does not use the rule it is trying to prove. **For #7, just fill in the blanks with the correct justification of each step.**

Prove that $a^3 \cdot a^5 = a^{(3+5)}$

$a^3 \cdot a^5 = (aaa) \cdot (aaaaa)$	Def. of exponents
$= (aaaaaaaa)$	Associative law
$= a^8$	Definition of exponents
$= a^{(3+5)}$	Substitution using $8 = 3 + 5$

□

#8 Prove that $(a^3)^5 = a^{3 \cdot 5}$

We will prove this directly. We have:

$$(a^3)^5 = (aaa)^5$$

definition of exponents

$$= (aaa) \cdot (aaa) \cdot (aaa) \cdot (aaa) \cdot (aaa)$$

definition of exponents (again)

$$= aaaaaaaaaaaaaa$$

associative law of multiplication

$$= a^{15}$$

Definition of exponents

$$= a^{3 \cdot 5}$$

Substitution using $15 = 3 \cdot 5$

□

#9 Prove that $a^5 \cdot b^5 = (ab)^5$

We will prove this statement directly. So, we have:

$(ab)^5 = (aaaaa) \cdot b^5$	definition of an exponent (on a)
$= (aaaaa) \cdot (bbbbb)$	definition of an exponent (on b)
$= aaaaabbbbb$	by multiplication
$= ababababab$	by the law of commutativity
$= (ab) \cdot (ab) \cdot (ab) \cdot (ab) \cdot (ab)$	by the law of associativity
$= (ab)^5$	definition of an exponent

□

#10 Considering the above, when we add the definition of exponents to the algebra, are the “rules” of exponents really a new idea or just a consequence of the existing structure of algebra? Give a thoughtful and complete answer.

Rules of exponents for what we have covered so far are just a consequence of the existing structure of algebra, or even just shorthand when you need to show a mathematical expression multiplied a given number of times. Generally, from the proofs on #7, #8, and #9, all of the steps boil down to multiplication laws, associativity, and commutativity. I do feel this only holds true if you have a^n where n is a natural number or positive integer. When looking at a^n when n becomes negative, rational, or even irrational, the idea of an exponent and proofs with exponents get much more complex and nontrivial. However, with the help of expanded theorems and the definition of logarithms, proofs of more complex exponent statements become more feasible versus only working with basic mathematical and logical axioms.

Notes – ungraded, add as many pages as you like.

Use these questions or other notes to help you organize your study and prepare for the exam.

1. Which ideas were new to you?
2. What questions might make good exam questions?
3. What ideas will you want to review later?
4. What were some “a-ah” moments, or big ideas you want to remember?